power — bring-up pass table

Run this board alone on the current-limited bench supply. Tick each box and write the reading. Any FAIL stops the sequence; fix it before moving on.

1. Before power (multimeter, supply OFF, USB unplugged)

☐ Every part is seated, pin 1 matches the drawing, no solder bridge between neighbouring pads.

☐ No dead short: GND to each supply rail reads at least 1 kΩ and is stable (record: _____).

☐ Rail to rail (12 V, 5 V, 3.3 V nets) reads at least 1 kΩ.

☐ Bare ground bars (4): each bar touches only its own pads; continuity along each bar below 1 Ω; no bar touches a signal pad.

☐ Every removable link is in its documented state (table 4).

2. First power-up

☐ Set the bench supply to 12 V with a current limit of 100 mA.

☐ Switch on. If the limit is reached, switch off at once and look for a short; do not raise the limit.

☐ Record the idle current: _____ mA.

☐ Adjust LM2596 to 5.00 V and XL4015 to 3.30 V with the consumer links (JP_LOGIC, JP_3V3) open; keep the 5 V buck below the guard trip (about 5.05 V).

☐ Fit the consumer links one at a time and re-check the rail LEDs.

3. Status LEDs

	LED	Lights when	From
☐	LED_12VFUSED	12 V fused input present	CP09
☐	LED_BUCK5V	5 V buck output present	CP09
☐	LED_BUCK3V3	3.3 V buck output present	CP09
☐	LED_5VSYS	5 V system rail present	CP09
☐	LED_3V3SYS	3.3 V system rail present	CP09
☐	LED_NOTFAULT	no fault latched	CP11

4. Removable links (default: shunt fitted)

	Link	Net	Open means
☐	JP_R7_PASS	ALLOW_RAIL	Open: the interlock logic is cut off from the student-rail switch; measure ALLOW_RAIL with the rail provably OFF.
☐	JP_R7_ENABLE	EDU_PWR_EN	Open: the interlock AND gate ignores the router enable; drive its input by hand to test the gate.
☐	JP_R7_ALIVE	ROUTER_ALIVE	Open: the interlock sees the watchdog as stopped; apply 3.3 V to the far pad to simulate a live heartbeat.

Every link opens to OFF: the loaded side is pulled low by an existing or added resistor.

power — test points and expected readings

Pads are existing pads; solder a short tin loop on the pad if you want a clip point. Readings are with the bench supply on and the student rail OFF.

	Test point	Pad	Column, row	Install	Expected	Tool	Reading
☐	TP_GND_1	J_PROBE.1	6, 1	CP09	0 V reference (left side)	DMM	_____
☐	TP_GND_2	U_POWER_INTERLOCK_ALLOW.13	73, 6	CP09	0 V reference (right side)	DMM	_____
☐	TP_12V_FUSED	J_POWER_INPUT_SW_MAIN.1	30, 35	CP09	12.0 V (after fuse and reverse diode)	DMM	_____
☐	TP_BUCK_5V	J_PROBE.5	10, 1	CP09	5.00 V, below the guard trip (about 5.05 V)	DMM	_____
☐	TP_BUCK_3V3	J_PROBE.6	11, 1	CP09	3.30 V	DMM	_____
☐	TP_12V_SYS	J_PROBE.4	9, 1	CP09	12.0 V once the consumer link is fitted	DMM	_____
☐	TP_5V_SYS	J_PROBE.2	7, 1	CP09	5.00 V once JP_LOGIC is fitted	DMM	_____
☐	TP_3V3_SYS	J_PROBE.3	8, 1	CP09	3.30 V once JP_3V3 is fitted	DMM	_____
☐	TP_5V_EDU	J_POWER_INTERLOCK_MATRIX.1	45, 5	CP09	0 V while OFF; at most 5.25 V when ON	DMM + scope	_____
☐	TP_EDU_PWR_EN	J_PROBE.7	12, 1	CP09	0 V at boot and reset; 3.3 V only while enabled	DMM	_____
☐	TP_ALLOW_RAIL	JP_R7_PASS.1	68, 34	CP11	0 V while OFF (enable AND no fault AND alive)	DMM	_____
☐	TP_NOT_FAULT	U_POWER_INTERLOCK_ALLOW.2	70, 6	CP11	high when no fault latched (verify polarity on the schematic)	DMM	_____
☐	TP_PRESET_N	J_PROBE.8	13, 1	CP09	high when idle, pulled low by a trip (verify on the schematic)	Scope	_____
☐	TP_ROUTER_ALIVE	J_POWER_INTERLOCK_MATRIX.9	49, 5	CP09	high while the heartbeat is fresh; low within 350 ms of HB OFF	Scope	_____

Polarity and idle levels marked “verify” must be confirmed against the schematic before they are used as pass criteria. A generated table is not measured evidence.

router-matrix — bring-up pass table

Run this board alone on the current-limited bench supply. Tick each box and write the reading. Any FAIL stops the sequence; fix it before moving on.

1. Before power (multimeter, supply OFF, USB unplugged)

☐ Every part is seated, pin 1 matches the drawing, no solder bridge between neighbouring pads.

☐ No dead short: GND to each supply rail reads at least 1 kΩ and is stable (record: _____).

☐ Rail to rail (12 V, 5 V, 3.3 V nets) reads at least 1 kΩ.

☐ Bare ground bars (4): each bar touches only its own pads; continuity along each bar below 1 Ω; no bar touches a signal pad.

☐ Every removable link is in its documented state (table 4).

2. First power-up

☐ Set the bench supply to the USB or 5 V source with a current limit of 100 mA.

☐ Switch on. If the limit is reached, switch off at once and look for a short; do not raise the limit.

☐ Record the idle current: _____ mA.

☐ With JP_R7_EN open, plug the ESP1 USB. The board must power up without enabling the student rail.

☐ Load the bring-up firmware and run the self-test sequence in selftest.md (PING, VER, RAILS, FAULT?, I2C SCAN, HB ON/OFF).

3. Status LEDs

	LED	Lights when	From
☐	LED_USBROUTER3V3	router 3.3 V present	CP00
☐	LED_5VSYS	5 V system rail present	CP11
☐	LED_12VSYS	12 V system rail present	CP11
☐	LED_ROUTERALIVE	watchdog output high (heartbeat fresh)	CP11

4. Removable links (default: shunt fitted)

	Link	Net	Open means
☐	JP_R7_STBA	STROBE_A	Open: matrix A strobe is cut off from the control buffer; test the buffer and the MT8816 separately.
☐	JP_R7_STBB	STROBE_B	Open: matrix B strobe is cut off from the control buffer.
☐	JP_R7_EN	EDU_PWR_EN	Open: ESP1 cannot enable the student rail whatever the firmware does; use for the first power-up and for bring-up of the ESP.
☐	JP_R7_RST	RST_RELEASE	Open: the matrix reset can never be released by ESP1; use to test the reset chain with the matrix held in reset.

Every link opens to OFF: the loaded side is pulled low by an existing or added resistor.

router-matrix — test points and expected readings

Pads are existing pads; solder a short tin loop on the pad if you want a clip point. Readings are with the bench supply on and the student rail OFF.

	Test point	Pad	Column, row	Install	Expected	Tool	Reading
☐	TP_GND_1	J_PROBE.1	6, 1	CP00	0 V reference (left side)	DMM	_____
☐	TP_GND_2	U_MATRIX_CONTROL_DIAG.5	75, 7	CP00	0 V reference (right side)	DMM	_____
☐	TP_USB_ROUTER_3V3	J_PROBE.3	8, 1	CP00	3.3 V from the router USB	DMM	_____
☐	TP_5V_SYS	J_PROBE.2	7, 1	CP00	5.00 V	DMM	_____
☐	TP_12V_SYS	J_PROBE.4	9, 1	CP00	12.0 V	DMM	_____
☐	TP_EDU_PWR_EN	J_ROUTER_CORE_PROTOTYPE.4	20, 20	CP00	0 V at boot and reset	DMM	_____
☐	TP_HEARTBEAT	J_PROBE.6	11, 1	CP00	50 ms edges only while the session is fresh	Scope	_____
☐	TP_ROUTER_ALIVE	J_PROBE.5	10, 1	CP00	high while the heartbeat is fresh	Scope	_____
☐	TP_EDU_FAULT	J_PROBE.8	13, 1	CP00	low when clear (verify polarity)	DMM	_____
☐	TP_XBAR_DAT	U_ROUTER_CORE_ESP1.21	14, 6	CP00	0 V idle, shift data when configuring	Scope	_____
☐	TP_XBAR_CLK	U_ROUTER_CORE_ESP1.28	14, 13	CP00	0 V idle, shift clock when configuring	Scope	_____
☐	TP_XBAR_LATCH	J_PROBE.7	12, 1	CP00	0 V idle, latch pulse after a shift	Scope	_____
☐	TP_STB_A_IN	U_ROUTER_CORE_ESP1.27	14, 12	CP00	0 V idle	Scope	_____
☐	TP_STB_B_IN	U_ROUTER_CORE_ESP1.25	14, 10	CP00	0 V idle	Scope	_____
☐	TP_STROBE_A	JP_R7_STBA.1	46, 33	CP24	0 V idle, strobe pulse on a write (50 ms or less)	Scope	_____
☐	TP_STROBE_B	JP_R7_STBB.1	50, 33	CP24	0 V idle, strobe pulse on a write (50 ms or less)	Scope	_____
☐	TP_XBAR_RESET	U_MATRIX_SWITCHES_MT_A.3	38, 6	CP24	held in reset until RESET_RELEASE and ALIVE	Scope	_____
☐	TP_RST_RELEASE	JP_R7_RST.1	8, 35	CP00	0 V at boot	DMM	_____
☐	TP_FIX_SDA	J_ROUTER_CORE_FIXTURE.3	6, 27	CP00	3.3 V idle (pull-up)	Scope	_____
☐	TP_FIX_SCL	J_ROUTER_CORE_FIXTURE.4	7, 27	CP00	3.3 V idle (pull-up)	Scope	_____

Polarity and idle levels marked “verify” must be confirmed against the schematic before they are used as pass criteria. A generated table is not measured evidence.